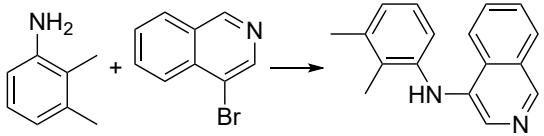
a C–N (Pd/Cu/Ni)

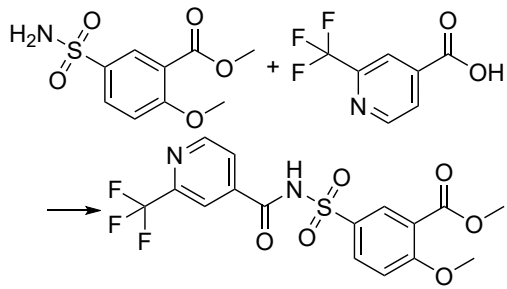
100 °C, 16 h



hit (C151)	$Pd_2(dba)_3$, BrettPhos, $Zn(OAc)_2$, DME	**25.0%**
added (C160)	**$Pd(OAc)_2$**, **XPhos**, **NaOtBu**, **t-AmOH**	**95.0%**

b Amide coupling

DMF, 25 °C, 2 h

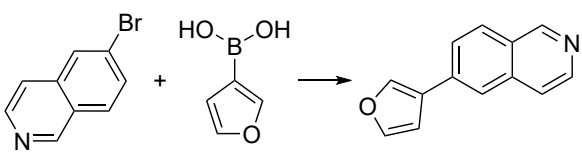


hit (C37)	PyOxim, DIPEA	**29.7%**
added (C31)	**DCC**, **NHS**, DIPEA	**76.9%**

library best 78.6%

c Suzuki (Pd)

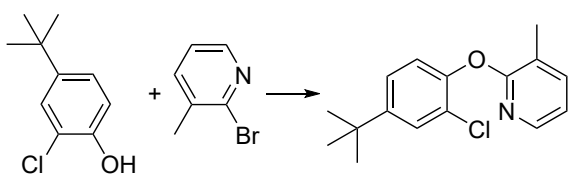
1,4-dioxane/H_2O (4:1), 100 °C, 12 h



hit (C35)	XPhos Pd G3, Cs_2CO_3	**54.2%**
added (C38)	**$Pd(dba)_2$**, **$P(tBu)_3$**, **K_3PO_4**	**95.0%**

d C–O (Ullmann library)

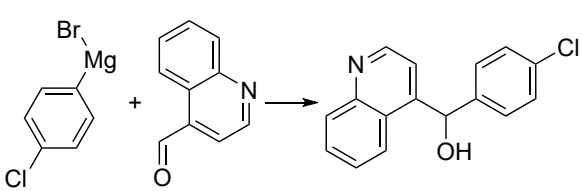
1,4-dioxane, 130 °C, 16 h



hit (C55)	CuI, TAEA, KOtBu	**45.4%**
added (C76)	**$Pd(OAc)_2$**, **XantPhos**, **K_2CO_3**	**85.5%**

e Grignard addition

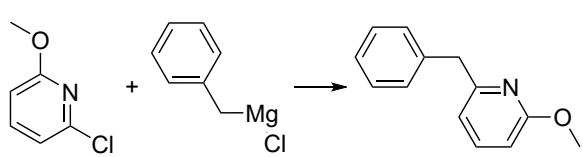
25 °C, 3 h



hit (C15)	LiCl, THF	**55.0%**
added (C39)	**$ZnBr_2$**, **CPME**	**95.0%**

f Kumada (Pd/Ni/Fe)

25 °C, 2 h



hit (C96)	$NiCl_2(dppp)$, THF/Et_2O (1:1)	**61.1%**
added (C46)	**$Ni(dppf)Cl_2$**, **KI**, **THF**	**98.3%**

g Recipe changes in the same improved cases

